Wheat

This section presents information on water relations and water management of wheat and provides links to other sources of information.

Crop Description and Climate

Bread and durum wheat (Triticum aestivum and T. turgidum) were domesticated in the Near and Middle East. Present world production is about 582.7 million tons from 213.8 million ha.(FAOSTAT, 2001).

The crop is grown as a rainfed crop in the temperate climates, in the sub-tropics with winter rainfall, in the tropics near the equator, in the highlands with altitudes of more than 1500 m and in the tropics away from the equator where the rainy season is long and where the crop is grown as a winter crop.

Wheat is grown under irrigation in the tropics either in the highlands near the equator and in the lowlands away from the equator. In the subtropics with summer rainfall the crop is grown under irrigation in the winter months. In the subtropics with winter rainfall it is grown under supplemental irrigation.

The length of the total growing period of spring wheat ranges from 100 to 130 days while winter wheat needs about 180 to 250 days to mature. Daylength and temperature requirements are key factors in variety selection. Varieties can be grouped as winter or spring types according to chilling requirements, winter hardiness and daylength sensitivity. Winter wheat requires a cold period or chilling (vernalization) during early growth for normal heading under long days. Winter wheat in its early stages of development exhibits a strong resistance to frost, down to - 20°C. The resistance is lost in the active growth period in spring and during head development and flowering periods frost may lead to head sterility. Because of this sometimes more damage is done to the winter crop by spring frost than by winter frost.

In areas of severe winters, cold winds and little snow, spring wheat varieties are grown. Spring wheat does not require chilling for heading and it is day-neutral. However, it is also sensitive to frost. For winter and spring wheat minimum daily temperature for measurable growth is about 5°C. Mean daily temperature for optimum growth and tillering is between 15 and 20°C. Occurrence of (spring) frost is an important factor in selection of sowing date. A dry, warm ripening period of 18°C or more is preferred. Mean daily temperatures of less than 10 to 12°C during the growing season make wheat a hazardous crop. Knowledge of genetic characteristics and particularly the growth and development pattern of wheat varieties is essential for meeting the combination of various climatic requirements for growth development and yield formation.

The crop can be grown on a wide range of soils but medium textures are preferred. Peaty soils containing high sodium, magnesium or iron should be avoided. The optimum pH ranges from 6 to 8. For good yields the fertilizer requirements are up to 150 kg/ha N, 35 to 45 kg/ha P and 25 to 50 kg/ha K.

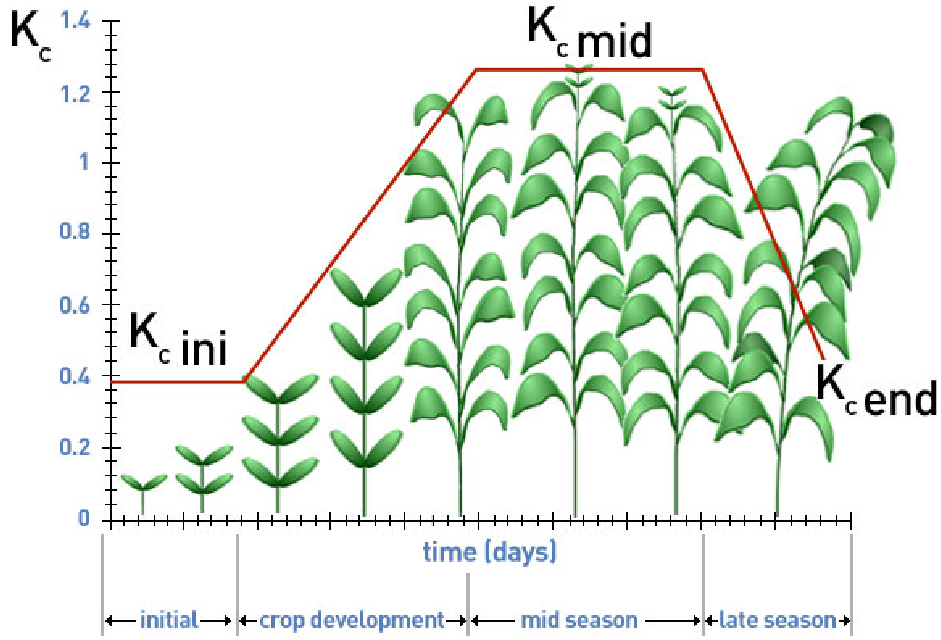
Wheat is relatively tolerant to a high groundwater table; for sandy loam to silt loam a depth of groundwater of 0.6 to 0.8 m can usually be tolerated, and for clay 0.8 to 1 m. For short periods the crop can withstand without visible harm a minimum depth of 0.25 m. With a rise of groundwater table to 0.5 m for long periods the yield decrease is 20 to 40 percent.

The crop is moderately tolerant to soil salinity but the ECe should not exceed 4 mmhos/cm in the upper soil layer during germination. Yield decrease due to salinity is 0% at ECe 6.0, 10/ at 7.4, 25% at 9. 5, 50% at 13 and 100% at ECe 20 mmhos/cm.

With pre-irrigation or sufficient rain to wet the upper soil layer, seeds are drilled 2 to 4 cm deep. as against 5 to 8 cm in dry soils, so that light showers will not cause the seeds to germinate. Under favourable water supply including irrigation and adequate fertilization row spacing is 0.12 to 0.15 m (450 to 700000 plants/ha) but increases to 0.25 m or more under poor rainfall conditions (less than 200000 plants/ha). Sowing rates under irrigation are 100 to 120 kg/ha (drilled) to 110 to 140 kg/ha (broadcast). Wheat is often grown in rotation and legumes, sunflower and maize are considered as suitable rotation crops.

The graph below depicts the crop stages of wheat, and the table summarises the main crop coefficients used for water management.

- CropWat
- CLIMWAT
- Diagnostic Tools for Investment
- FAO Nile
- ETo Calculator
- GAEZ
- GeoNetwork
- GLADIS
- HWSD
- Rapid Appraisal Procedure (RAP)
- SoiLEX
- SoilSTAT
- WaPOR
- WaterLex



	Stages of Development					Plant date	Region
Crop characteristic	**Initial**	**Crop Development**	**Mid-season**	**Late**	**Total**		
Stage length,							
days	20	25	60	30	135	Mar/Apr	35 - 45°L
Wheat	15	30	65	40	150	July	East Africa
	40	30	40	20	130	April	
	40	60	60	40	200	Nov	
	20	50	60	30	160	Dec	Calf.Desert,USA
	20[1]	60[1]	70	30	180	Dec	Calf.,USA
Winter Wheat	30	140	40	30	240	Nov	Mediterranean
	160	75	75	25	335	Oct.	Idaho, USA
Depletion Coefficient, p							
Spring Wheat	0.5	>>	0.5	0.8	0.55		
Winter Wheat	0.6	>>	0.6	0.9	0.55		
Root Depth, m							
Spring Wheat	0.3	>>	>>	1.2	-		
Winter Wheat	0.3	>>	>>	1.4	-		
Crop					-		
Coefficient,							
Kc	0.3	>>	1.15	0.25-0.40[2]			
Spring Wheat	0.4	>>	1.15	0.25-0.40[2]			
Winter Wheat							
with	0.7	>>	1.15	0.25-0.40[2]			
- frozen soils							
-non-frozen soils							
Yield Response Factor, Ky							
Spring Wheat	0.2	0.65	0.55	-	1.05		
Winter Wheat	0.2	0.6	0.5	-	1.15		

[1]These periods of winter wheat with lengthen in frozen climates according days having zero growth potential and wheat dormancy. Under general conditions and in the absence of local data , fall planting of winter wheat can be presumed to occur in northern temperate climates when the 10-day running average of mean daily air temperature decreases to 17°C or Decenmber 1, whichever comes first. Planting of spring wheat can be presumed to occur when the 10-day running average of mean daily air temperature increases to 5°C. Spring planting of maize-grain can be pressure.

[2]The higher value is for hand-harvested crops.

Water Requirements

Water Supply And Crop Yield

Water Uptake

Irrigation Scheduling

Selected Recommendations for Different Regions

Yield